Exploring Polynomial Manipulations in Maxima

Maxima is a computer algebra system based on Macsyma and written in Lisp. This 16th article in the series, 'A Mathematical Journey Through Open Source', demonstrates polynomial manipulations using Maxima.

Polynomials have fascinated mathematicians for ages because of the wide variety of their applications, ranging from basic algebra and puzzles to the various sciences. We are going to look at some of the polynomial manipulation functions provided by Maxima, and use some of them for a couple of real world applications.

Fundamental polynomial operations

Let's start with a demonstration of the fundamental polynomial operations, like addition, subtraction, multiplication and division. In all these, whenever needed, we'll use *expand()* to expand the polynomials, and *string()* to display the polynomials in a flattened notation.

```
$ maxima -q
(%i1) p1: x^2 - y^2 + y$
(%i2) p2: -x^2 - y^2 + x$
(%i3) p3: (x + y)^2$
(%i4) string(p1 + p2);
(%o4)                                   -2*y^2+y+x
(%i5) string(p1 + p2 + p3);
```